FINAL PROJECT REPORT

Global AI Trends in Education Across Different Countries

Project Tool: Tableau Public

Project Area: Data Engineering / Data Visualization

Submitted by: Pranita Prashant Kadam

Class / Division: A

Roll No.: 29

Academic Project Report

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1. Introduction

1.1 Project Overview

Artificial Intelligence (AI) is increasingly being used in education to support learning, teaching, content creation, assessment, and educational planning. This project uses Tableau Public to explore and present patterns in AI adoption and usage across selected countries and regions. The visual analytics project is titled “Global AI Trends in Education Across Different Countries.”

The workbook contains multiple worksheets and Tableau story points. The screenshots show a country-level comparison, a map of selected countries, a comparison of student and teacher AI usage, an adoption journey over time, rural-versus-urban usage, a regional comparison, and a breakdown of adoption levels. These views are brought together to make the information easier to understand.

1.2 Objectives

• To present global AI-in-education information in a clear visual format.

• To compare average AI usage across selected countries.

• To compare student and teacher AI usage measures.

• To show trends over time for school, student, and teacher AI adoption or usage.

• To compare rural and urban AI usage patterns.

• To classify records into high, moderate, and low adoption levels.

• To create a Tableau dashboard and story that help viewers explore the data.

2. Project Initialization and Planning Phase

2.1 Define Problem Statement

Information about AI use in education can be difficult to interpret when it is stored only in rows and columns. A viewer may struggle to compare countries, understand changes over time, or identify differences between student, teacher, rural, and urban usage. The problem addressed by this project is how to transform the dataset into an interactive and understandable visual report.

2.2 Project Proposal (Proposed Solution)

The proposed solution is to use Tableau Public to connect to the education dataset, review and prepare the available fields, create calculated fields where required, build individual worksheets, and combine the worksheets into dashboards and a story. Color, labels, maps, bars, lines, and a donut chart are used to present different aspects of the dataset.

2.3 Initial Project Planning

• Understand the dataset fields and the meaning of each measure.

• Check data types, missing values, duplicate records, and inconsistent category names.

• Create calculated fields needed for adoption categories or percentage measures.

• Build individual worksheets for geographic, comparative, trend, and category analysis.

• Arrange worksheets in dashboards and add titles and legends.

• Create story points to guide the viewer through the main findings.

• Review labels, scales, colors, readability, and dashboard layout before submission.

3. Data Collection and Preprocessing Phase

3.1 Data Collection Plan and Raw Data Sources Identified

The project screenshots identify the workbook/data source as “Global AI in Education.” The precise original publisher, download URL, license, and collection date are not visible in the screenshots and should be added by the project author from the actual source file or download page. The visible fields and chart labels indicate country, region, year, school/student/teacher usage or adoption measures, rural and urban usage, AI tools, and adoption category.

3.2 Data Quality Report

A formal quality report should be completed by checking the source data directly. Based on the visual workbook, the following checks are recommended. Do not report a check as passed until it has been verified against the source data.

• Completeness: identify blank values in country, region, year, and measure columns.

• Validity: confirm that year fields are valid years and usage/adoption measures use the intended numeric type.

• Consistency: standardize country names and region labels, including any shortened names used in chart labels.

• Duplicates: check whether repeated rows represent valid separate observations or duplicate records.

• Range checks: investigate negative percentages, percentages above 100 where not expected, and unusually large values.

• Category checks: confirm that adoption labels are consistently written as High Adoption, Moderate Adoption, and Low Adoption.

3.3 Data Exploration and Preprocessing

The visual exploration stage uses country names and geographic fields to plot selected countries on a world map, compares measures across countries, and places year on a time axis for trend analysis. In Tableau, preprocessing may include changing data types, renaming fields, setting geographic roles, formatting measures as percentages, creating calculated fields, and checking aggregation levels.

A calculated field called “Adoption cal field” appears in the workbook legend. The screenshots show three categories: High Adoption, Low Adoption, and Moderate Adoption. The exact formula is not visible, so it should be documented from the calculated-field editor rather than guessed in this report.

4. Data Visualization

4.1 Framing Business Questions

• Which selected countries have higher or lower average daily AI usage?

• How do student AI usage and teacher AI usage compare across countries?

• How have school, student, and teacher AI adoption/usage measures changed across the years shown?

• How do rural and urban AI usage patterns compare over time?

• How do school AI adoption measures compare across world regions?

• How many records fall into high, moderate, and low adoption categories?

• Which AI tools appear in the dataset, and what share of usage is associated with each tool?

4.2 Developing Visualizations

The workbook uses a range of chart types so that each question can be answered with a suitable visual. Bar charts support country comparisons; a line chart supports time trends; a world map shows geographic distribution; a mirrored horizontal bar view compares rural and urban patterns; a donut chart compares regions; and a categorical bar chart summarizes adoption levels. Titles, labels, legends, and consistent colors help viewers interpret each chart.

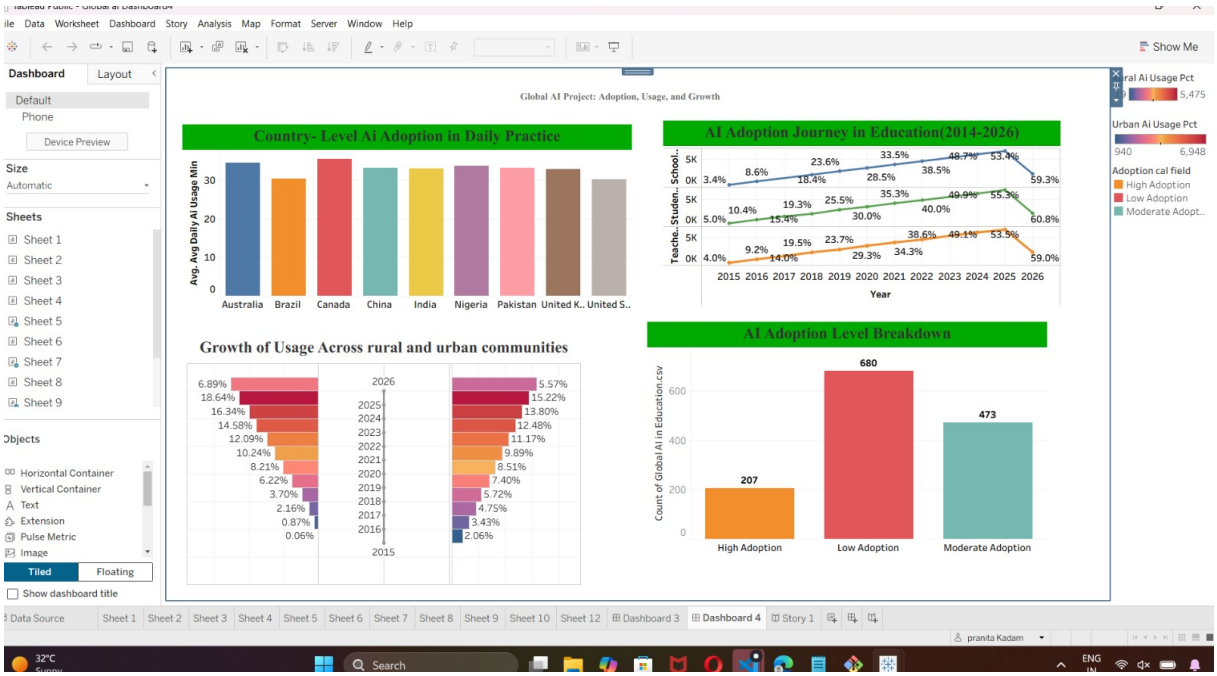


Figure 1. Dashboard showing country-level daily usage, adoption journey, rural/urban comparison, and adoption-level breakdown.

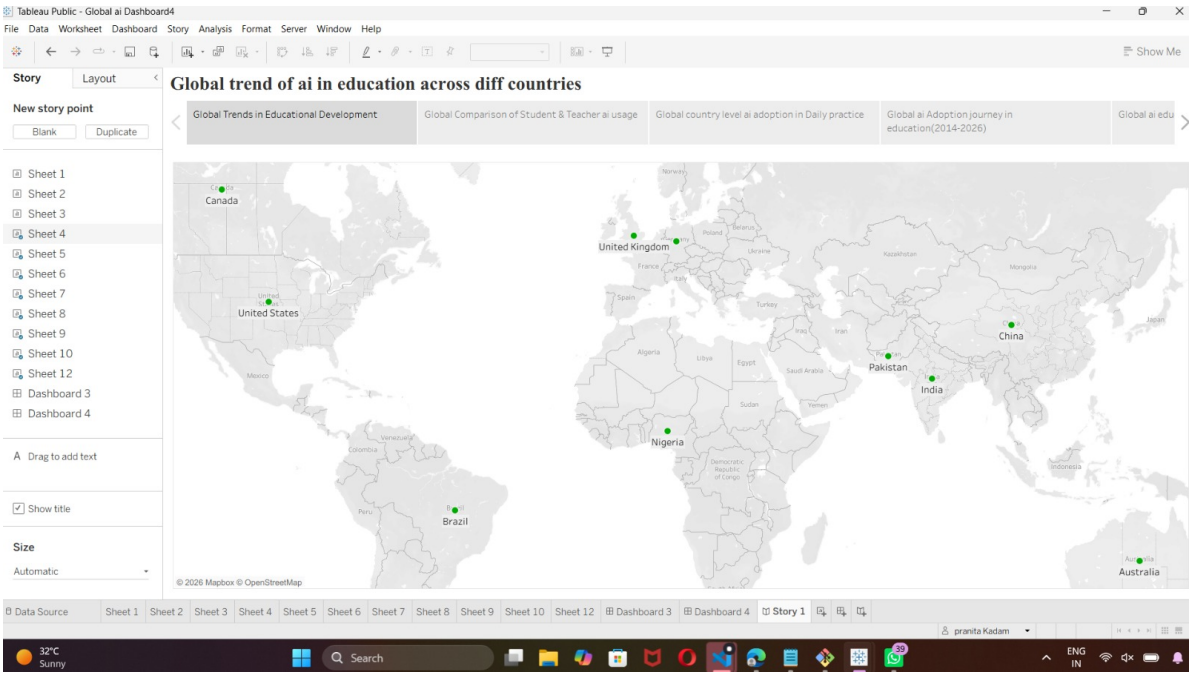


Figure 2. Tableau story point showing geographic distribution across selected countries.

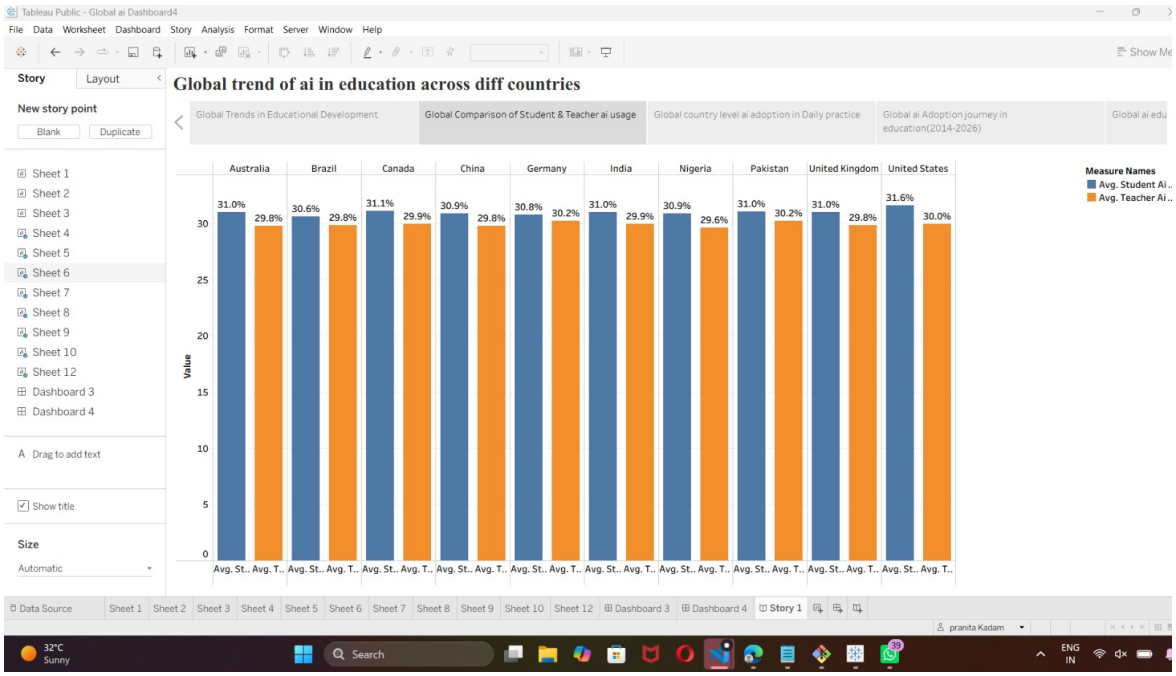


Figure 3. Tableau story point comparing average student and teacher AI usage across countries.

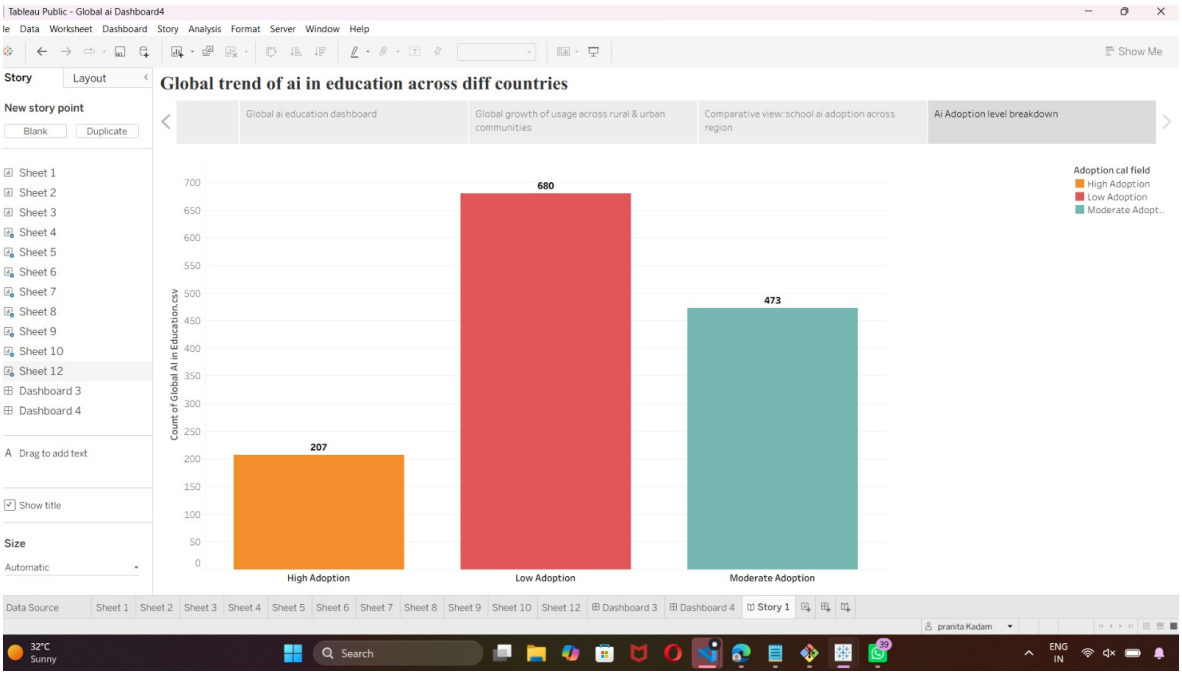


Figure 4. Tableau story point showing the adoption-level breakdown.

5. Dashboard

5.1 Dashboard Design File

The Tableau workbook includes dashboard views that arrange multiple worksheets on a single canvas. One dashboard, titled “Global AI Project: Adoption, Usage, and Growth,” combines country-level average daily usage, a multi-series adoption journey from 2014–2026, rural and urban usage comparisons, and an adoption-level breakdown. Another dashboard screenshot includes a geographic map, student/teacher comparison, AI tool comparison, and regional school-adoption view.

The visible design uses green title bands, white chart backgrounds, colored marks, and legends. This creates a recognizable theme. To improve presentation quality, use consistent capitalization, correct spelling in chart titles, avoid clipped labels, and ensure that axes and percentage labels match the underlying measures. Check that

every dashboard object fits without overlapping.

6. Report

The report component communicates the main patterns revealed by the visualizations. In the country comparison screenshot, average daily AI usage differs by country, with Canada and Australia appearing among the higher bars and Brazil and the United States among the lower bars in that view. Student usage bars appear slightly higher than teacher usage bars in the comparison chart for the countries shown.

The adoption journey chart displays an upward trend through 2025 in the plotted series, followed by a visible drop in 2026. This should be interpreted carefully: the last-year decrease may reflect incomplete-year data, a change in data collection, or a genuine change. The dataset and calculation should be checked before drawing a firm conclusion.

The adoption-level chart displays 207 records for High Adoption, 680 for Low Adoption, and 473 for Moderate Adoption. These are counts shown in the screenshot, not necessarily counts of unique schools or countries; the exact unit depends on the source table.

6.1 Story Design File

The Tableau Story is titled “Global trend of AI in education across diff countries.” Story points guide viewers through the geographic overview, student and teacher comparison, country-level daily usage, adoption journey, AI-tool comparison, rural/urban growth, regional comparison, and adoption-level breakdown. This sequence helps the audience move from a broad overview to more specific comparisons.

For the final submission, save the workbook with clearly named worksheets and story points. Confirm that each story point has a meaningful title and that charts are readable in presentation mode.

7. Performance Testing

7.1 Utilization of Data Filters

The screenshots show legends and multiple story points, but they do not clearly confirm which interactive filters are configured. Verify the dashboard by testing any country, region, year, AI-tool, or adoption-category filters that have been added. Confirm that selections update the intended worksheets and that clearing a filter restores the full view.

7.2 Number of Calculated Fields

At least one calculated field is visible: “Adoption cal field.” The screenshots are not sufficient to determine the total number of calculated fields in the workbook. Open the Data pane and inspect calculated fields to record the exact number. Document each field name, formula, and purpose.

7.3 Number of Visualizations

The screenshots show at least seven distinct visualization types or analytical views: a world map, country comparison bar chart, student/teacher comparison bar chart, adoption journey line chart, rural/urban mirrored bar chart, regional donut chart, AI-tool comparison chart, and adoption-level bar chart. Some are repeated across dashboards and story points. Count the worksheets in the workbook to report the exact total.

Test	Expected result
Open each dashboard/story point	All views load without errors.
Check titles and labels	Titles are readable and labels are not clipped.
Check legends and colors	Each category matches its intended color and legend.
Test filters (if present)	Relevant views update correctly; reset restores all data.
Check calculated field	Adoption categories match the documented formula.

Test	Expected result
Check trend chart	Year order and aggregation are correct; 2026 is validated.
Save and reopen workbook	Dashboards and story points remain intact.

8. Conclusion / Observation

This project demonstrates how Tableau Public can turn education-related AI data into an interactive visual story. The charts allow users to compare countries, inspect changes over time, compare student and teacher usage, view rural/urban patterns, and examine adoption categories. The screenshots suggest that student usage is slightly higher than teacher usage in the countries shown and that the plotted time series generally increases through 2025.

The dashboard also shows more records in the Low Adoption category than in the High Adoption category. These observations are descriptive and depend on the dataset, its definitions, and the aggregation used in Tableau. The project is useful for visual exploration, but any policy or performance conclusions should be supported by verified source documentation and data-quality checks.

9. Future Scope

• Add verified source citations, dataset version, collection date, and a data dictionary.

• Add interactive filters for country, region, year, AI tool, and adoption category.

• Include tooltips with definitions, record counts, and source notes.

• Validate the 2026 values and explain whether the year contains complete data.

• Add statistical summaries and comparisons across income groups or education levels if suitable fields exist.

• Improve accessibility with readable font sizes, color-blind-friendly palettes, and descriptive titles.

• Publish the Tableau workbook or packaged workbook and provide a shareable demo link.

• Refresh the dashboard periodically when a new dataset version becomes available.

10. Appendix

10.1 Source Code (if any)

This project is primarily implemented in Tableau Public using worksheets, dashboards, story points, and a calculated field. No separate programming source code is visible in the supplied screenshots. Add the exact calculated-field formula(s) in this section by opening each calculated field in Tableau and copying its formula.

10.2 GitHub & Project Demo Link

GitHub repository: Not provided.

Tableau Public project/demo link: Not provided.

Before submission, paste the actual repository and published Tableau URL here if available. Do not submit placeholder links as working links.

Submission Checklist

• Attach the original dataset or cite its exact source.

• Verify the data quality report against the raw data.

• Record the exact number of calculated fields and worksheets.

- Confirm filter behavior and dashboard/story navigation.
- Replace blank student, class, roll number, and guide details on the cover page.
- Add the Tableau Public demo URL and GitHub URL if available.